TM294

industrialPhysics

Requirements

Necessary basic knowledge	Create structured applications in Automation Studio.
Training modules	TM210 - Working with Automation Studio
Software	industrialPhysics 2.4.x Automation Studio 4.3 and later
Hardware	None

Table of contents

1 Introduction

Automation Studio offers end-to-end simulation that allows applications to be configured and tested without requiring hardware. By importing CAD data into **industrialPhysics**, digital twins of plants can be created and then connected to Automation Studio.
By creating a communication task in industrialPhysics, existing Automation Studio projects can be connected to the simulated systems.

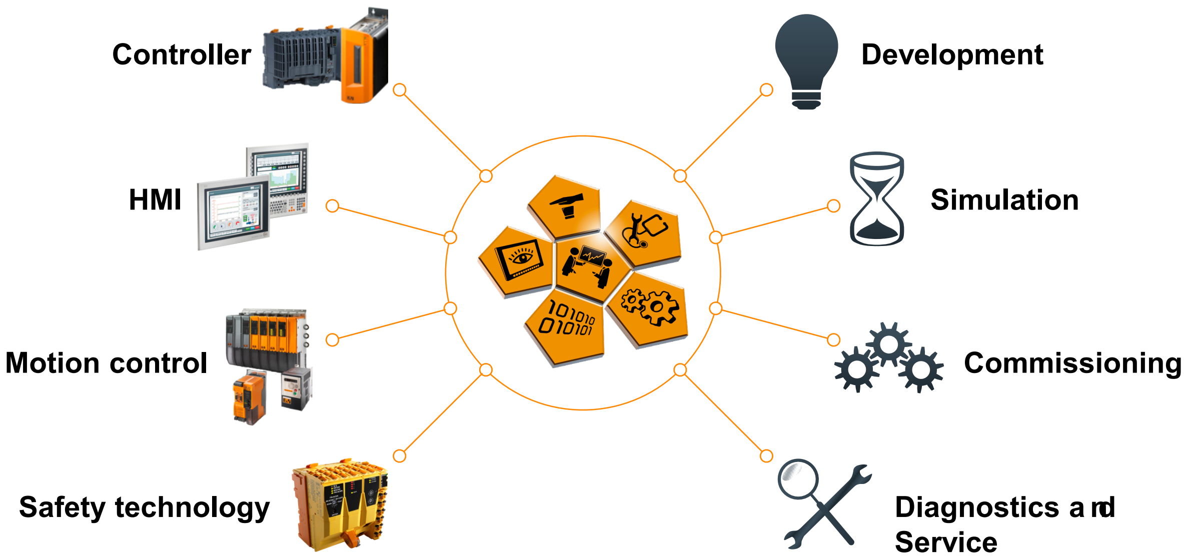


Figure 1: Automation Studio: One engineering tool for the machine's entire lifecycle

1.1 Learning objectives

This training module gives participants insight into the basics of simulation in industrialPhysics and how to handle the interface to Automation Studio.

- Participants will get an overview of terminology related to virtual commissioning and simulation.
- Participants will receive information about the development process ranging from the model to simulation and virtual commissioning.
- Participants will get an overview of the different levels of simulation and the respective areas of application.
- Participants will learn how to work with industrialPhysics.
- Participants will learn how to make individual elements in an industrialPhysics project dynamic.
- Participants will learn how to create projects in industrialPhysics.
- Participants will learn how to establish an online connection between an industrialPhysics model and an Automation Studio project.

1.2 Symbols and safety notices

Unless otherwise specified, the symbol descriptions and safety notices listed in "TM210 – Working with Automation Studio" apply.

2 General information

The sharp increase in production requirements, the expanding complexity of systems and the development of greater numbers of technical products with software all make simulation an essential component of the development process.
The following image shows the typical B&R simulation levels, which differ in their level of detail. A distinction is made between the automation hardware level, the component and machine level and the process and plant level.
Starting at the base, you have simulation of hardware and software components. Automation Studio supports all possible simulation options.
Dynamic processes of machines and components can be simulated on the second level. For this, B&R offers connection to the most popular simulation tools on the market.
The third level enables simulation of complex system processes. For this type of simulation, B&R Automation Studio provides appropriate interfaces for external software such as industrialPhysics.

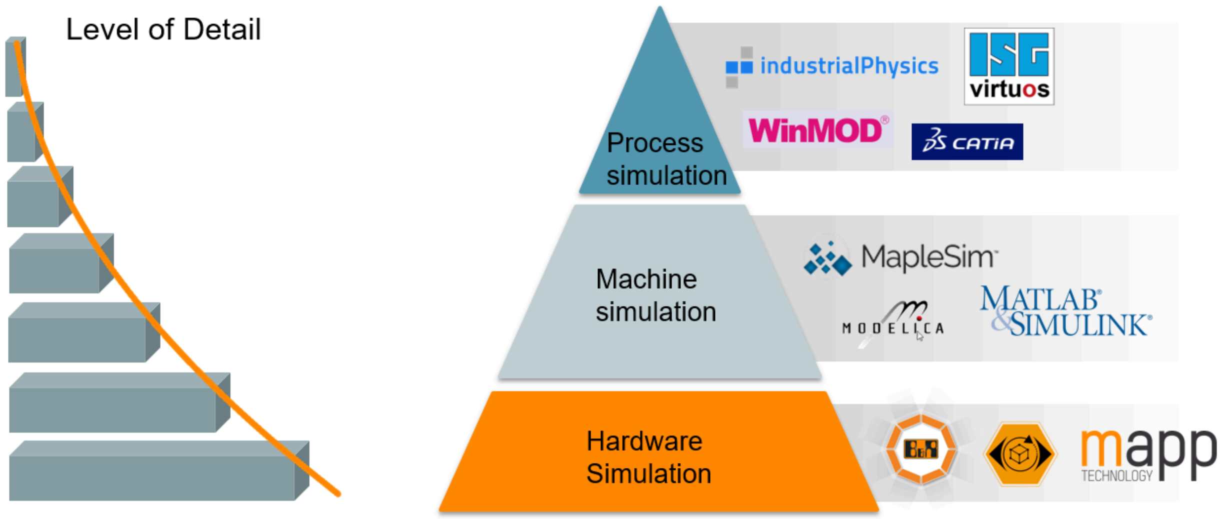


Figure 2: Simulation levels: Plants and processes, machines and components, hardware

2.1 Simulation

Simulation is an experimental approach to determine static and dynamic properties of a system based on a model. The model makes it possible to gain knowledge about the system and its behavior. This knowledge can then be transferred to reality. For the digital representation of the original system, it is irrelevant whether the system to be examined already exists or is still in planning stage. The image below shows a simplified procedure for creating a simulation model.

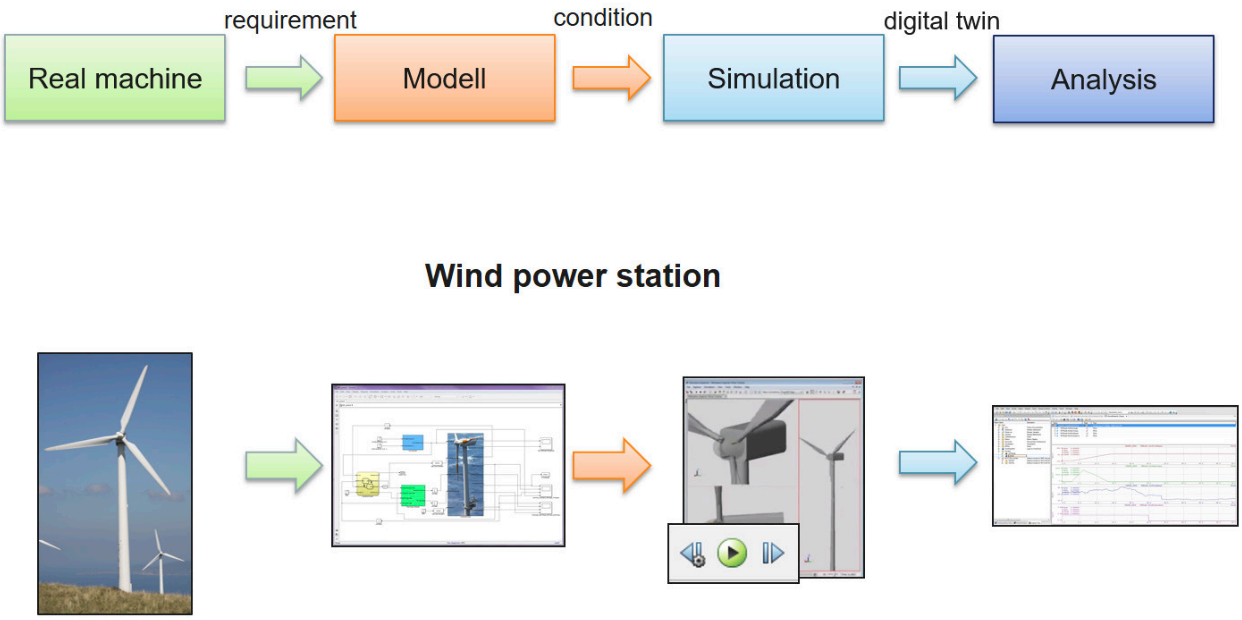


Figure 3: Simplified representation of a simulation.

Advantages industrialPhysics simulation with regard to software and system development

- Possibility to analyze extremely extensive or complicated systems
- Generation of highly efficient program code
- More precise adjustment when presetting system parameters
- Minimization and early detection of errors
- High efficiency and productivity
- High product quality
- Time and place for innovation
- Rapid prototyping
- Reusability
- Accelerated time to market

2.2 Model-based development

Model-based development is based the principle of simulation. A complex process is first represented in an abstract, realistic model. It is important that all processes necessary for development are contained in the simulation model since any further development is based on it. During the test phase, continuous improvements can be made based on the test results from the model. This process is displayed in the following diagram.

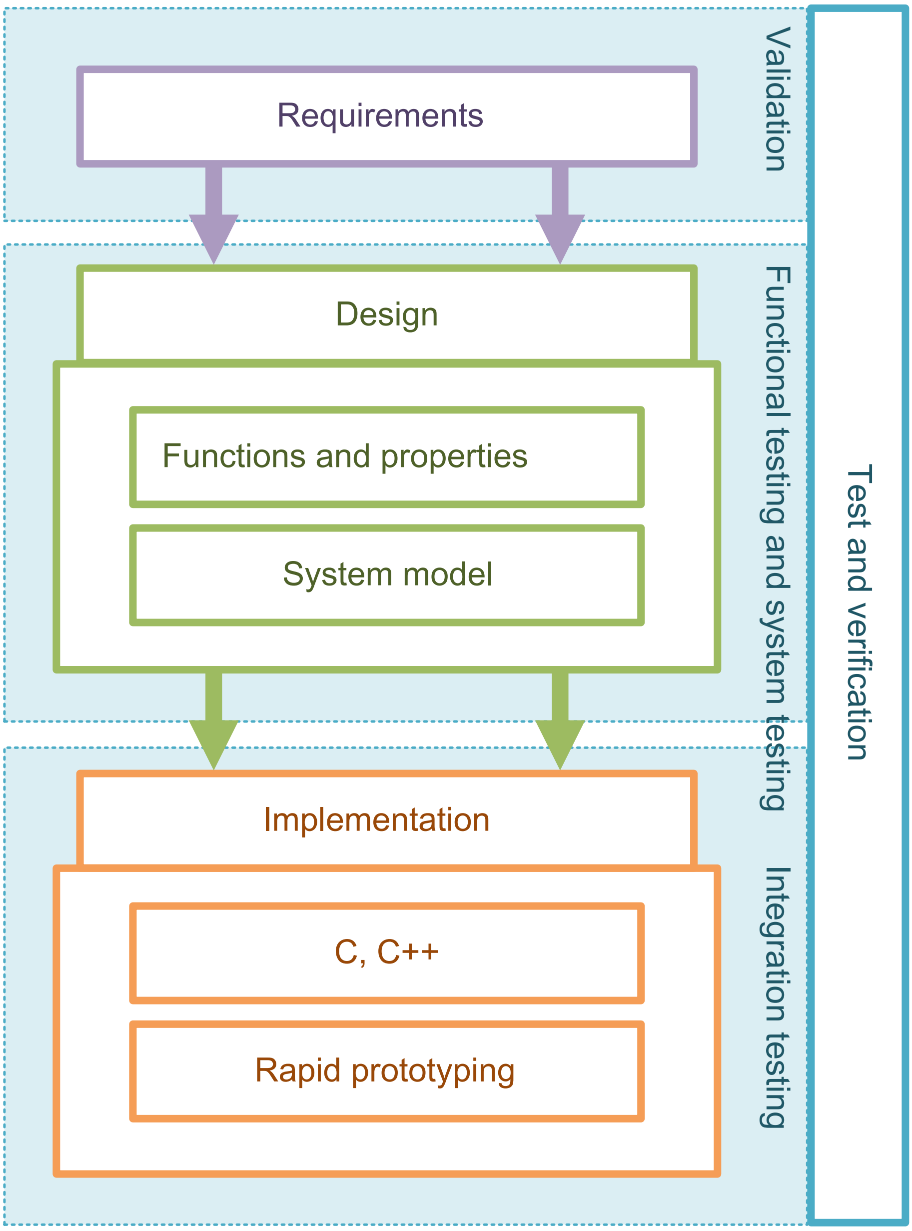


Figure 4: Procedure of model-based development

2.3 Virtual commissioning

Virtual commissioning refers to the testing of the simulation model in a virtual environment in order to simulate real commissioning as accurately as possible. In most cases of virtual commissioning, the simulation model is a virtual machine. Data is imported, tested and changed on the virtual machine before the software is transferred to the real machine. An executable model enables the software to be tested on a workstation computer or on a real-time system at an early stage.
An important aspect during virtual commissioning of mechanical processes is the 3D representation that simulates machine behavior and thus provides visual feedback for the tester.
It should be noted, however, that virtual commissioning does not replace real commissioning completely. In model-based development, two test procedures have proven to be effective: Software-in-the-loop (SiL) and hardware-in-the-loop (HiL).

2.3.1 Software-in-the-loop (SiL)

Software-in-the-loop is a test procedure used for models in a simulation environment. During a software-in-the-loop simulation, the software that has been developed and the simulation environment are executed on the same hardware. The software communicates with the simulation, which is running on the same processor, via global variables or variable mapping.

2.3.2 Hardware-in-the-loop (HiL)

Hardware-in-the-loop is a test procedure for evaluating a model on the real hardware used during commissioning. A real-time simulation is used for the test system in order to represent a controlled system for a closed-loop controller as accurately as possible. An HiL system provides the controller with all input and output signals that would exist in the real environment.

3 industrialPhysics

3.1 User interface

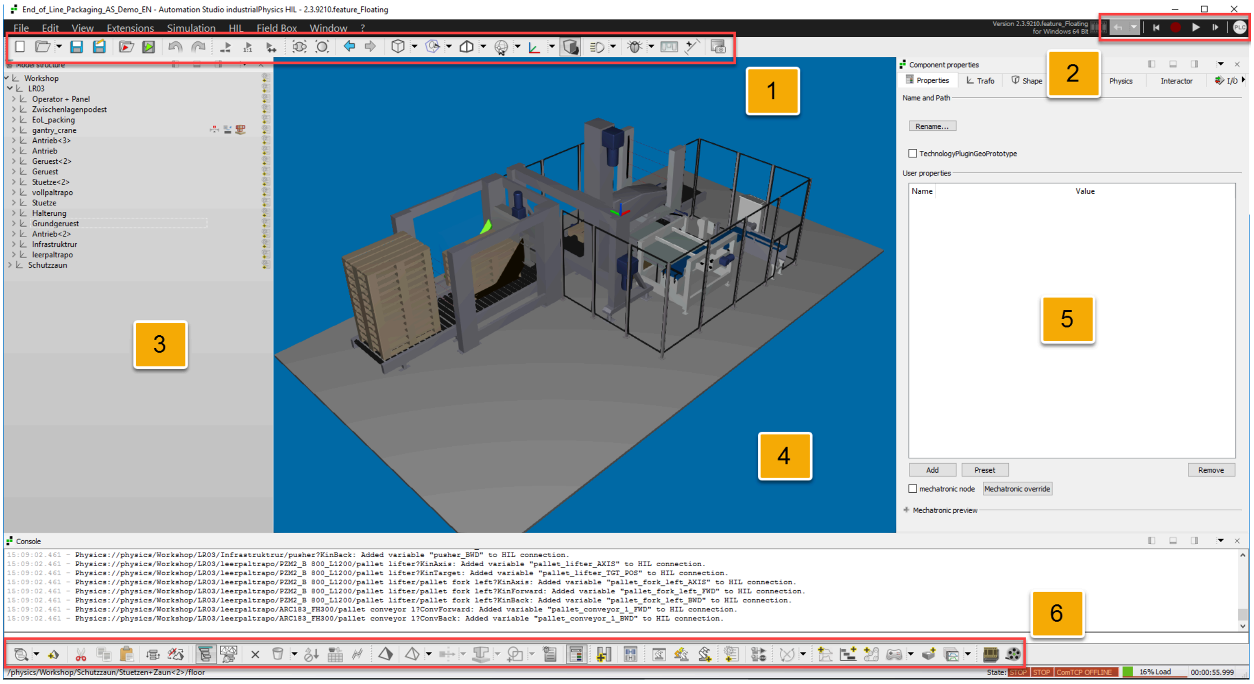


Figure 5: User interface

1) File toolbar

   3D toolbar

2) Simulation toolbar

3) Component tree

4) Model view

5) Properties window

6) Structure toolbar

   Component toolbar

   Joints

   Robots

   PLC development

   Kinematics

   Motion control

   Recording and playback

File toolbar

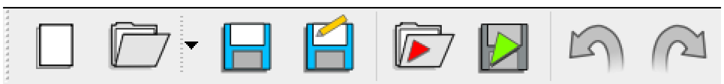


Figure 6: File toolbar

1) Create a new document; existing data is deleted
2) Open an existing document
3) Save the opened document
4) Save the opened document under a new name
5) Import a STEP document
6) Export a VRML document
7) One work step back
8) One work step forward

3D toolbar

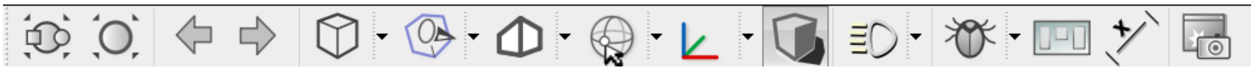


Figure 7: 3D toolbar

1) Zoom the whole model (Zoom Fit)
2) Zoom the selected component
3) Select a standard view
4) Edit user-defined view
5) Change between perspective and orthogonal view
6) Navigation, follow mode and SpacePilot
7) Choice of manipulator: None, coordinate, translation or rotation
8) Shadow on/off
9) Lighting management
10) Show physics information
11) Arrange windows automatically
12) Distance and angle measurement
13) Save 3D screenshot

Simulation toolbar

Figure 8: Simulation toolbar

1) Switch between simulation and playback
2) Create new simulation (easy or difficult reconstruction)
3) Reset simulation
4) Start the recording.
5) Starting the simulation
6) One simulation step forward
7) Switch on HiL mode

Structure toolbar

Figure 9: Structure toolbar

1) Search in model structure
2) Add new element
3) Cut element
4) Copy element
5) Add element
6) Drag-and-drop hierarchy manipulation on/off
7) Merge selected shapes
8) Show/Hide model structure
9) Reset the visibility of a component
10) Delete component
11) Bucket management
12) Suppress element on/off
13) Axis manager
14) Rail connections

Component toolbar

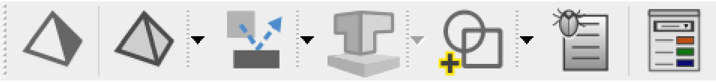


Figure 10: Component toolbar

1) Change component to "Invisible"
2) Physical property (immaterial, static, kinematic, dynamic or subcomponent)
3) Selection of collision type
4) Selection of collision body
5) Collision detection on/off
6) Debugging log on/off
7) Open component properties

Joints

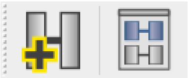


Figure 11: Joints

1) Create joint
2) Show joint connections

Robots

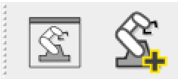


Figure 12: Robots

1) Robot control
2) Add robot kinematics

PLC development

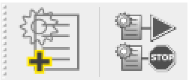


Figure 13: PLC development

1) Add script
2) Manage all model scripts

Kinematics

Figure 14: Kinematics

1) Add translational or rotational kinematics

Motion control

Figure 15: Motion control

1) Motion control
2) Add object for sequence control
3) Add chain
4) Add input device
5) Add user-defined motion control
6) Motion control management

Recording and playback

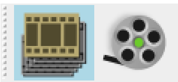


Figure 16: Recording and playback

1) Manage recorded simulation tracks
2) Create a video from the current simulation track

3.2 CAD import

Before a machine can be made dynamic, the CAD data must be imported into industrialPhysics.
This is done using one of the following two options:

Bidirectional CAD interface:

Depending on the license models of the following CAD programs, there is a toolbox with an interface that can be used to transfer the CAD data:

- AutoCAD Inventor
- CREO Parametrics
- IronCAD
- SolidEdge
- SolidWorks

Importing STEP data:

Almost any CAD program can export CAD data in STEP format. This format is standardized according to application log ISO 10303-2xx STEP.
Formats AP203 and AP214 were developed for mechanical CAD data. AP214 is the most recent format with more features and should therefore be used. AP214 is sometimes called "STEP with colors".

?/Knowledge center/doc/20_CAD/

3.3 Object properties

Physical properties

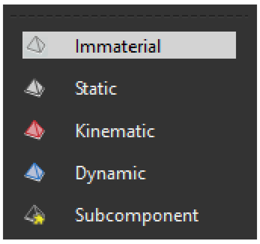


Figure 17: Physical properties

Immaterial: These components are excluded from any physical calculation (default setting). They have no weight and no collision body.

Static: Static objects have a collision body that does not move along with the object.

Kinematic: Kinematic objects behave like static objects, with the difference that the collision body moves along with the object.

Dynamic: Dynamic objects have a weight in addition to a collision body. They react to all physical events such as accelerations caused by collisions and gravity.

Subcomponent: A subcomponent inherits all the physical properties of the component above it.

Collision types

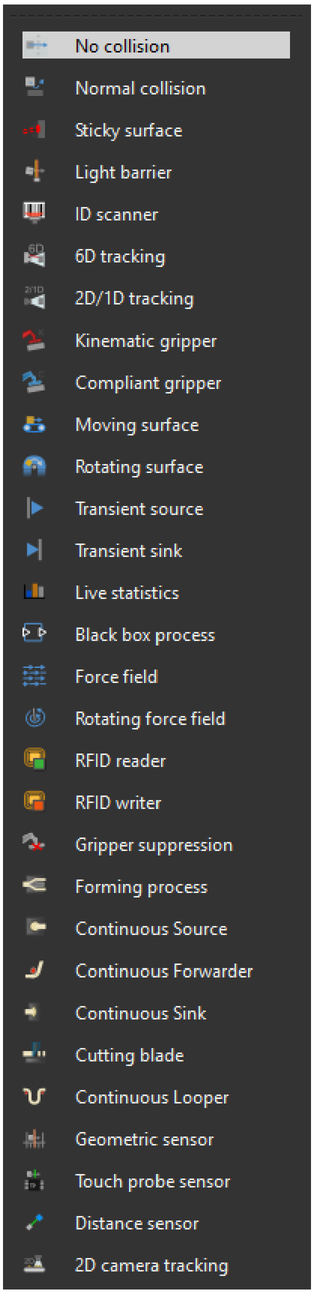


Figure 18: Collision types

No collision: Structure behaves as if it has no solid object geometry.
Normal collision: Default setting. A body behaves like an ideal solid object when colliding with dynamic objects.
Sticky surface: Dynamic objects adhere to the surface. Unlike grippers, they are permanent and cannot be deactivated.
Light barrier: During penetration with a dynamic body, one status output is set to TRUE.
ID scanner: Obsolete; replaced by RFID reader
6D tracking: Detects the position of a dynamic object as 6D transformer.
2D/1D tracking: Detects the position of a dynamic object as 3D vector.
Kinematic gripper: Permanently defines a maximum of one object at penetration Status returned as signal "Vacuum". Can be switched off.
Compliant gripper (dynamic): Permanently defines objects when penetrating at collision position or by aligning the transformers. Status returned as signal "Vacuum". Can be switched off.
Moving surface (linear conveyor): Dynamic objects are moved by friction in the direction of a specified vector.
Rotating surface (circular conveyor): Dynamic objects are moved by friction circularly around a specified vector.
Transient source: Generates transient objects based on a time or trigger.
Transient sink: Destroys transient objects during collision. Can be switched off.
Live statistics: Like IR light curtain. Additional counter for the total number of objects detected currently and during simulation runtime.
Black box process: Representation of a process as a black box with release time (cycle time), capacity and release direction. Captures dynamic objects at penetration, stacks them according to the direction defined if capacity is available and keeps them for the release time defined. If capacity is exceeded: Function "Normal collision" until capacity is available.
Force field: Dynamic objects are moved in the direction of a specified vector.
Rotating force field: Dynamic objects are moved circularly around a specified vector.

Collision types

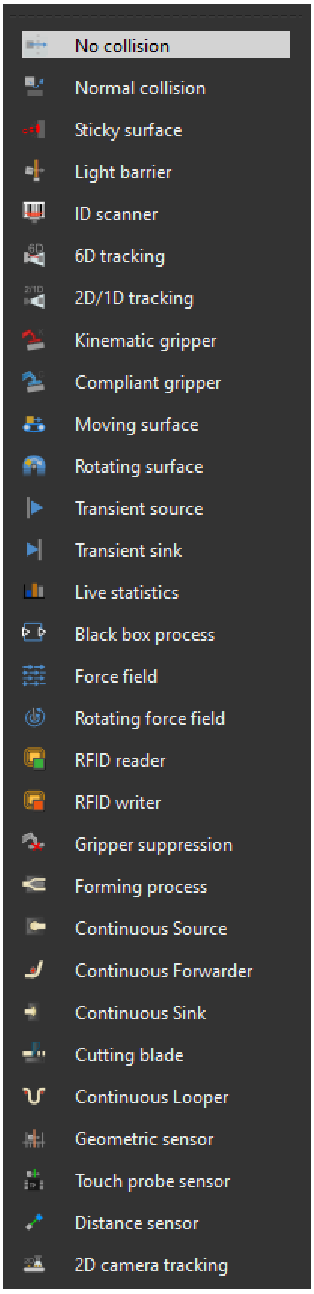


Figure 19: Collision types

RFID reader: Can read data from a dynamic object in the event of a collision.

RFID writer: Can read and write data from a dynamic object in the event of a collision.

Gripper suppression: Components gripped by a gripper are released at penetration with "Suppress gripper".

Forming process: Creates a continuous line of volumetric flow rate from an incoming component.
Cut a transient component using command "Cut".

Continuous source: Creates a continuous, transient line.
Cut a transient component using command "Cut".

Continuous forwarder: Conveys incoming transient components from a continuous source along the z-axis.

Continuous sink: Continuously destroys an incoming line of components from a continuous source.

Cutting blade: Splits objects at penetration on the plane that contains the origin of the blade and lies parallel to the x-y plane of the object to be split.
Cut only takes place at the edge of signal "Cut" during collision.

Continuous looper: Memory of a continuous product flow. Integrates via the difference between input and output speed and stores up to a defined capacity limit value.

Geometric sensor: Provides the minimum and maximum position of the object boundaries in relation to the z-axis of the sensor during penetration.

Touch probe sensor: Similar to a light barrier with an additional option of detecting the boundaries of the penetrating object along the z-axis of the sensor.

3.4 Knowledge center

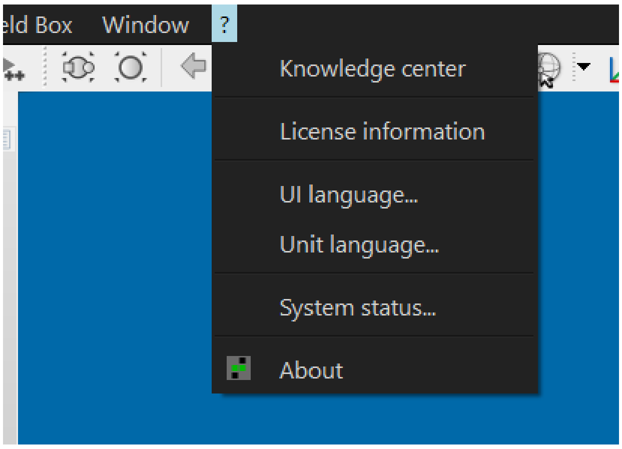


Figure 20: Knowledge center

The industrialPhysics knowledge center can be opened via the menu bar and contains several examples and documents for the various functions.

4 Working with industrialPhysics

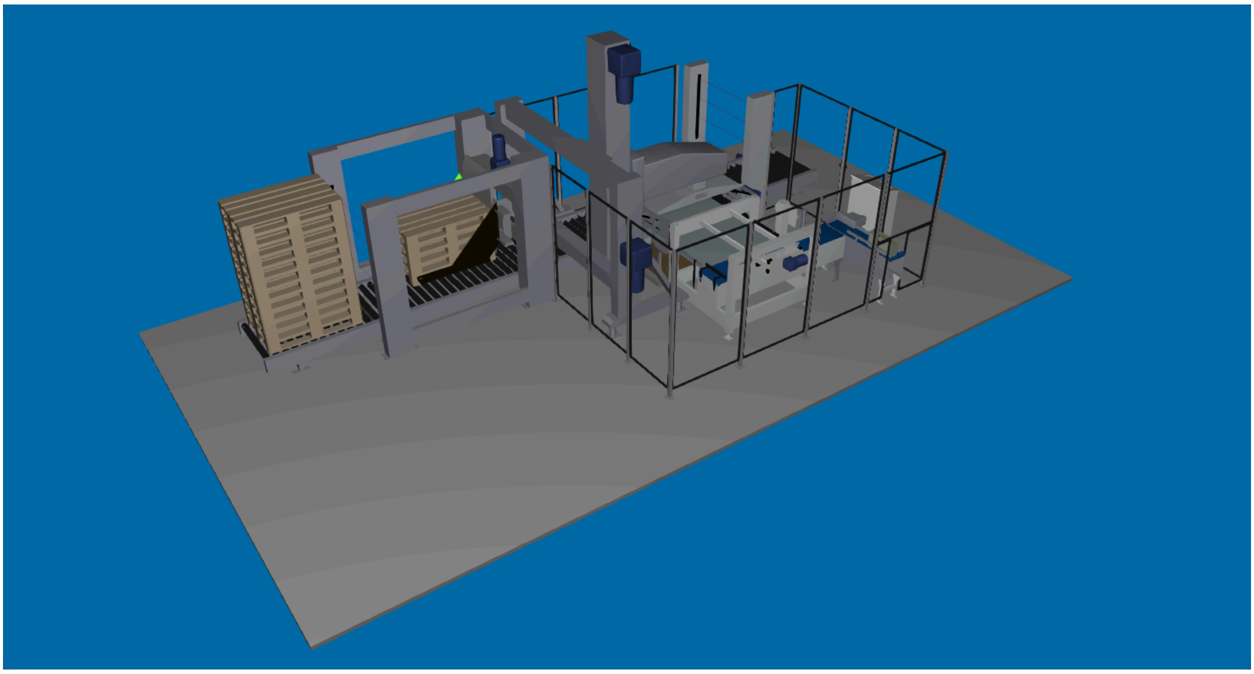


Figure 21: End-of-line example

The following examples are based on a palletizing machine for beverage cartons. 14 cartons are collected on a platform and then stacked into two layers on a pallet. When the pallet is full, it is passed on and a new empty pallet is lowered from a stack onto the conveyor belt.
In the next chapters, the following steps are carried out to make the machine completely dynamic:

- Activate the infeed conveyor belt
- Create a dynamic source from the product
- Activate the first pusher
- Commission the photoelectric sensors for packet detection
- Activate the two other conveyor belts
- Add a second photoelectric sensor
- Activating the rotary pusher
- Activate the carriage
- Activate the entire pallet truck including pusher and suction cup for cardboard
- Activate the pallet lift
- Commission the pallet conveyor belts for infeed and outfeed

4.1 First steps in industrialPhysics

Coordinate system

Each element in industrialPhysics has its own coordinate system. All movements, such as conveyor belt movements or pure translations, are performed based on this coordinate system. The directions are defined as follows:

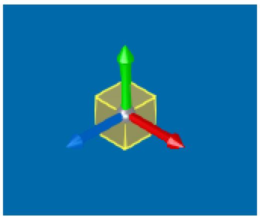


Figure 22: Coordinate system

X	Red
Y	Green
Z	Blue

To adjust the coordinate system of a body, first open the component window (ALT + SHIFT + M or right click in the model structure -> Component). By selecting **Modify transformations** under the **Transformation** tab, the coordinate system can be moved and rotated.

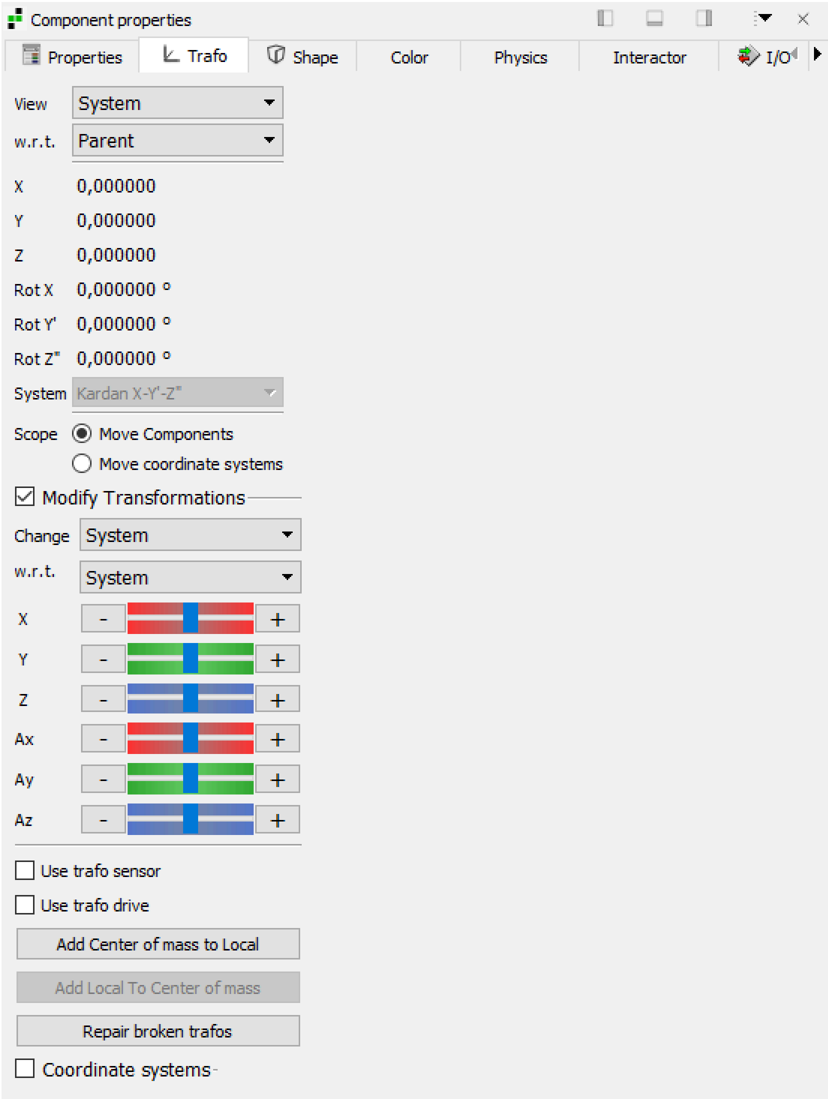


Figure 23: Component properties -> Transformation

Physical properties

As described in chapter 3.2, each component must be assigned one of the following properties.

Immaterial	Immaterial components cannot move, have no collision body and therefore cannot have a collision type. All imported components are immaterial by default.
Static	Static components have a collision body and can be assigned a collision type. The function of the collision body is only available at the original position of the component since the collision body of a static component does not move with it. (e.g. light barriers, conveyors, etc.)
Kinematic	Kinematic components are similar to static components, with the difference that their collision bodies move along. (e.g. gripper, pusher)
Dynamic	Dynamic components have the same properties as kinematic objects. In addition, dynamic components are affected by gravity and only these components can be detected by light barriers or moved by grippers. (e.g. cardboard boxes)

Table 1: Physical properties

Subcomponent	Subcomponents inherit all properties from the main component. (e.g. A cardboard box consists of cardboard and adhesive tape. A cardboard box is dynamic, cardboard and adhesive tape are subcomponents)

Table 1: Physical properties

Drive

A drive must be configured for many components (conveyor, force field, translation, rotation, etc.). There are a variety of drives, which are explained in the following table.

Direct drive	The component moves abruptly to a specified position. (v and a = ∞)
Speed drive	The component moves at a specified speed. (a = ∞)
Ramped drive	The component accelerates as configured at the specified speed.
Positioning drive	The component moves towards the specified position at the configured speed and acceleration.
MotionControl drive	Positioning drive with extended setting options via I/Os. (e.g. jerk time, etc.)
PID drive	Positioning drive that is controlled via a PID controller.
Vibration drive	The component vibrates at the specified amplitude and frequency.
Generic drive	Special drive used for TechnologyPlugins

Table 2: Drive

Conveyor

If a conveyor is used, the direction of movement must be defined in addition to selecting the drive. This is done by specifying a vector in the coordinate system of the object under **Direction** in the option menu.

> If the forward direction of a **conveyor** with **ramped drive** does not match the desired logical direction, it can be corrected without changing the coordinate system by entering a negative number.

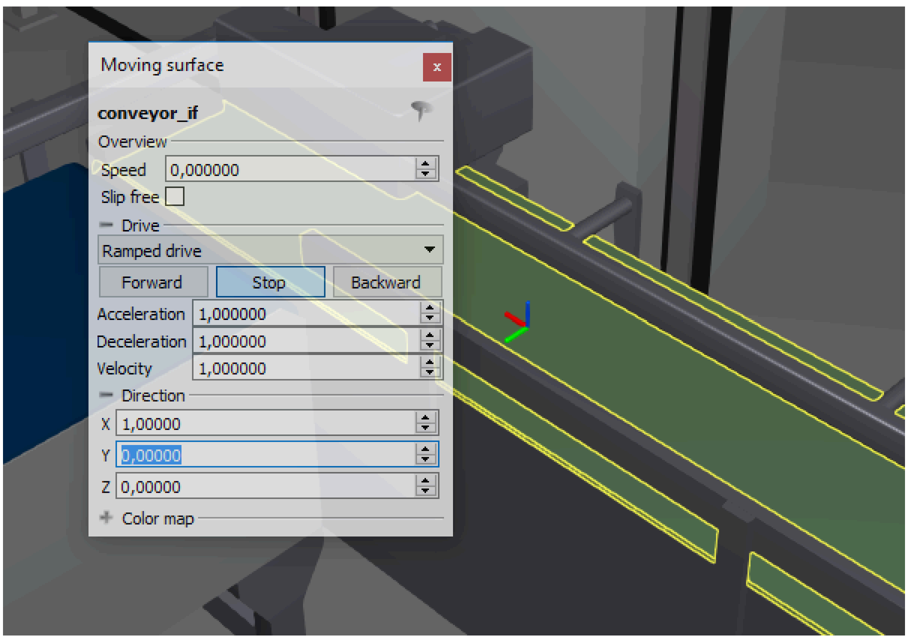


Figure 24: Conveyor - Direction

Source

During simulation, new objects can be created via sources. The new objects are created in certain time intervals or via a trigger variable based on a reference object. The newly generated objects are always **dynamic**, regardless of the physical properties of the source.
When a component is used as a source, the following properties can be configured.

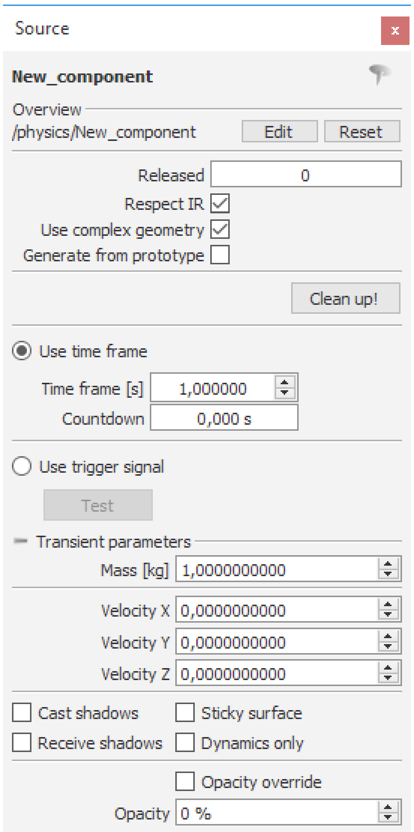


Figure 25: Source

Respect IR	Time-based or trigger-based releases are only performed if no other dynamic object is in the release area.
Use complex geometry	If the reference object consists of subcomponents, it is possible to decide whether the subcomponents or a simplified overall geometry should be created.
Generate from prototype	In order to generate dynamic components, a source must be created from object "NoShape" via the dynamic component and this option must be selected. The display of the prototype is suppressed and new instances are generated from the prototype instead. The source object is therefore no longer visible.
Use time frame / Use trigger signal	New objects are generated (time-controlled or trigger-controlled).
Transient parameters	Initial speed and properties with regard to graphical representation can be set for the newly generated object.

Table 3: Source settings

Task: Conveyor belt

This section shows how to open the previously described model and make its first components dynamic. The cardboard boxes are generated at the infeed conveyor belt and transported forward on the conveyor belt.
Adjust the properties of the required components and test the procedure on the first machine part.

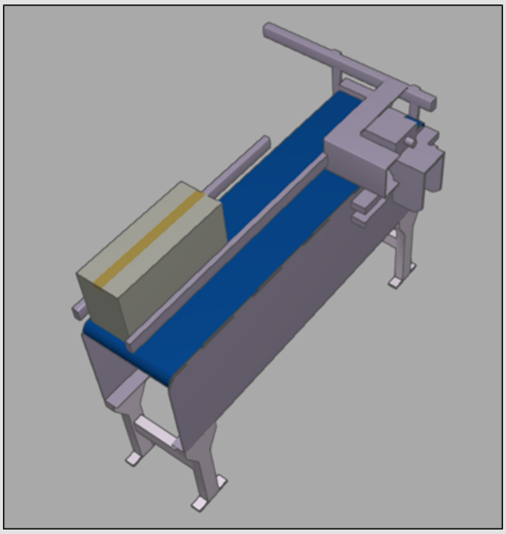


Figure 26: Infeed

1) Open iPhysics file **End_of_Line_Packaging_EN.iphx**.
2) Change the property of the **floor** to **Static**.
3) Change the property of the **box** to **Dynamic** and the property of the subelement to **Subcomponent**.
4) Start the simulation. What do you expect to happen?
5) Change the property of **Conveyor Belt_Infeed** to **Static** and select **Moving surface** as the collision type.
6) Adjust the conveyor. **(direction** and **drive**)
7) Create a **source** defined as the **box**.
8) Start the simulation. What do you expect to happen?

4.2 Light barrier and extended drive

Light barrier

Light barriers detect the penetration of a dynamic element into the collision body of the light barrier. The status is indicated by a BOOL variable. Another option is to decide whether this event should be displayed by the industrial-Physics light barrier. The signal can also be inverted.

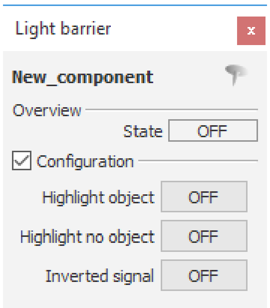


Figure 27: Light barrier

Extended drive

When drives are used for (translational) movement of objects, they must be configured with realistic limits.
It is also possible to scale the movement of a drive and define cam plates via this menu. These options are only needed if models are made dynamic entirely in industrialPhysics by scripts as opposed to controlling them via a PLC connection (as in this case).

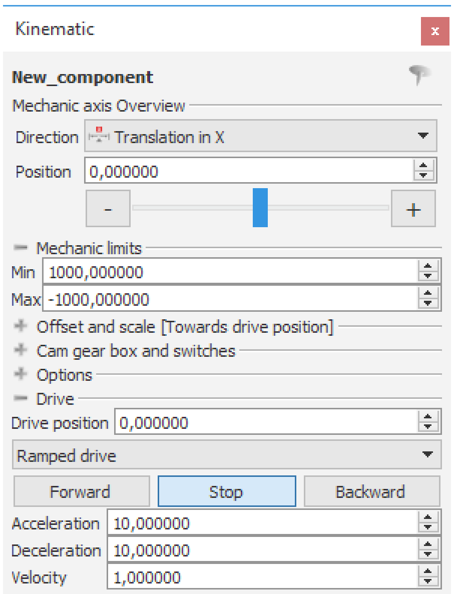


Figure 28: Drive - Limits

Buttons "- " and "+" can be used to change the position during active simulation. The limits determined this way must then be entered under **Mechanic limits**.

> If the forward direction of a translational movement does not match the desired logical direction, the local coordinate system of the object must be rotated. (see chapter 4.1 - Coordinate systems)

Task: Pusher

In this section, further components of the palletizing machine are put into operation.

To prevent the boxes from falling off the conveyor belt, the belts and limits must be set to "Static". It is necessary to detect when a box arrives at the end of the conveyor belt in order to trigger pushing it onto the next belt. Adjust the properties of the required components and test the entire procedure.

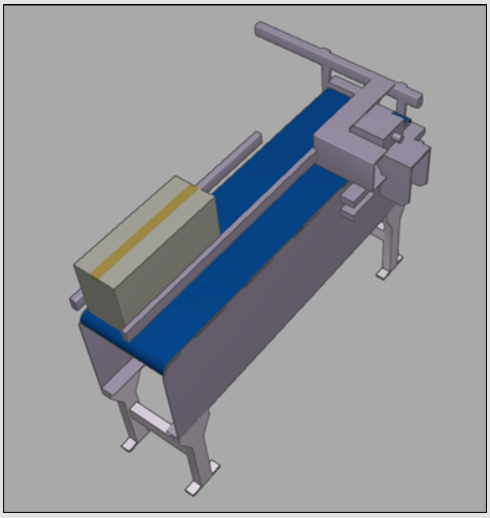


Figure 29: Infeed

1) Change the property of the **limit** and the **belts** to **Static**.
2) Change the property of the **sensor** to **Static** and the **collision type** to **Light barrier**.
3) Change the property of the **pusher** to **Kinematic** and activate **Translation in X.**
4) Adjust the **pusher**. **(drive with ramp** and **mechanical end stops**)
5) Start the simulation and test the functionality.

4.3 Connecting with Automation Studio

Configuration options

To establish a TCP connection with Automation Studio, it must first be configured in industrialPhysics. Different methods can be used for this, depending on the licenses. The HIL wizard is available with every full version. The industrialPhysics B&R runtime license does not include this feature. In this version, the TCP connection must be configured manually.

Manual configuration of the TCP connection

If an HIL connection does not yet exist in the project, it must first be created. This is done in the ComTCP view. (HIL -> ComTCP view or Alt+Shift+T)

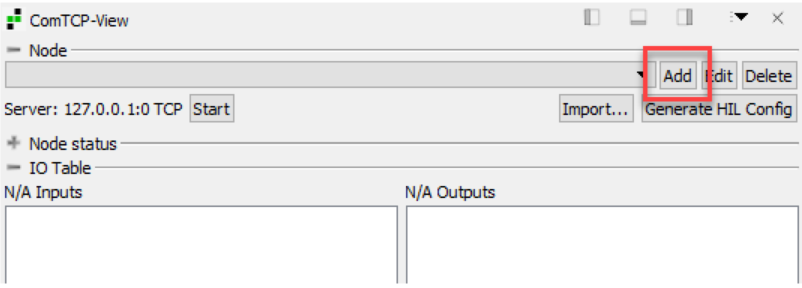


Figure 30: Adding the HIL connection

In the following window, it is necessary to specify a unique name, the functionality of industrialPhysics (TCP server or client), the address of the computer where the TCP server is running and the port.

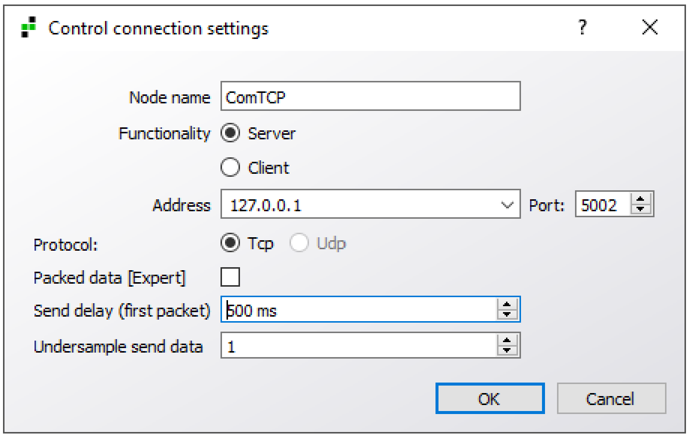


Figure 31: Settings for the HIL connection

In order to transfer inputs and outputs via TCP, they must be added to the HIL connection configuration just created by right-clicking in the I/O tab containing the components.

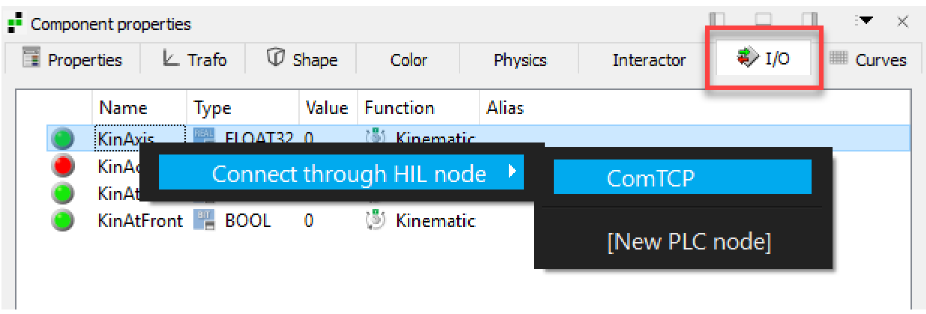
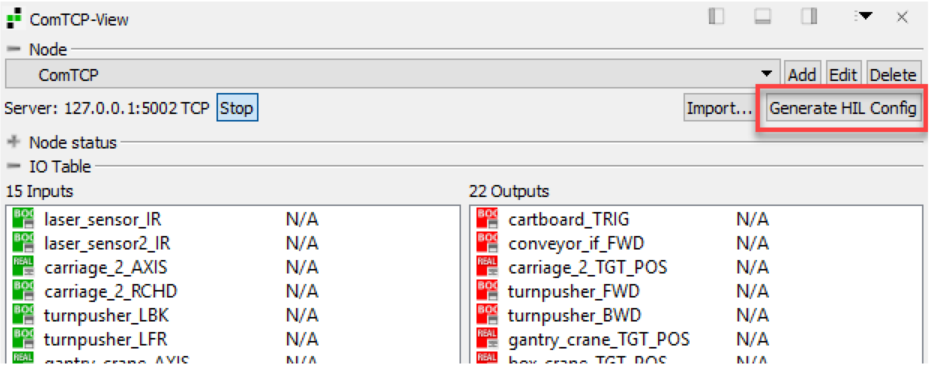


Figure 32: Adding variables to HIL connection configuration

When all inputs and outputs are defined, the packet must be generated and then imported into the AS project as described in Communication with Automation Studio.



HiL wizard

Automation Studio and industrialPhysics are connected via a communication task generated in industrialPhysics. This is done via menu option **HIL -> Start wizard**.
The connection must be given a unique name and the IP address of the computer where industrialPhysics is running must be entered. For ARsim: 127.0.0.1. Any port can be used. The package is generated in the specified directory.

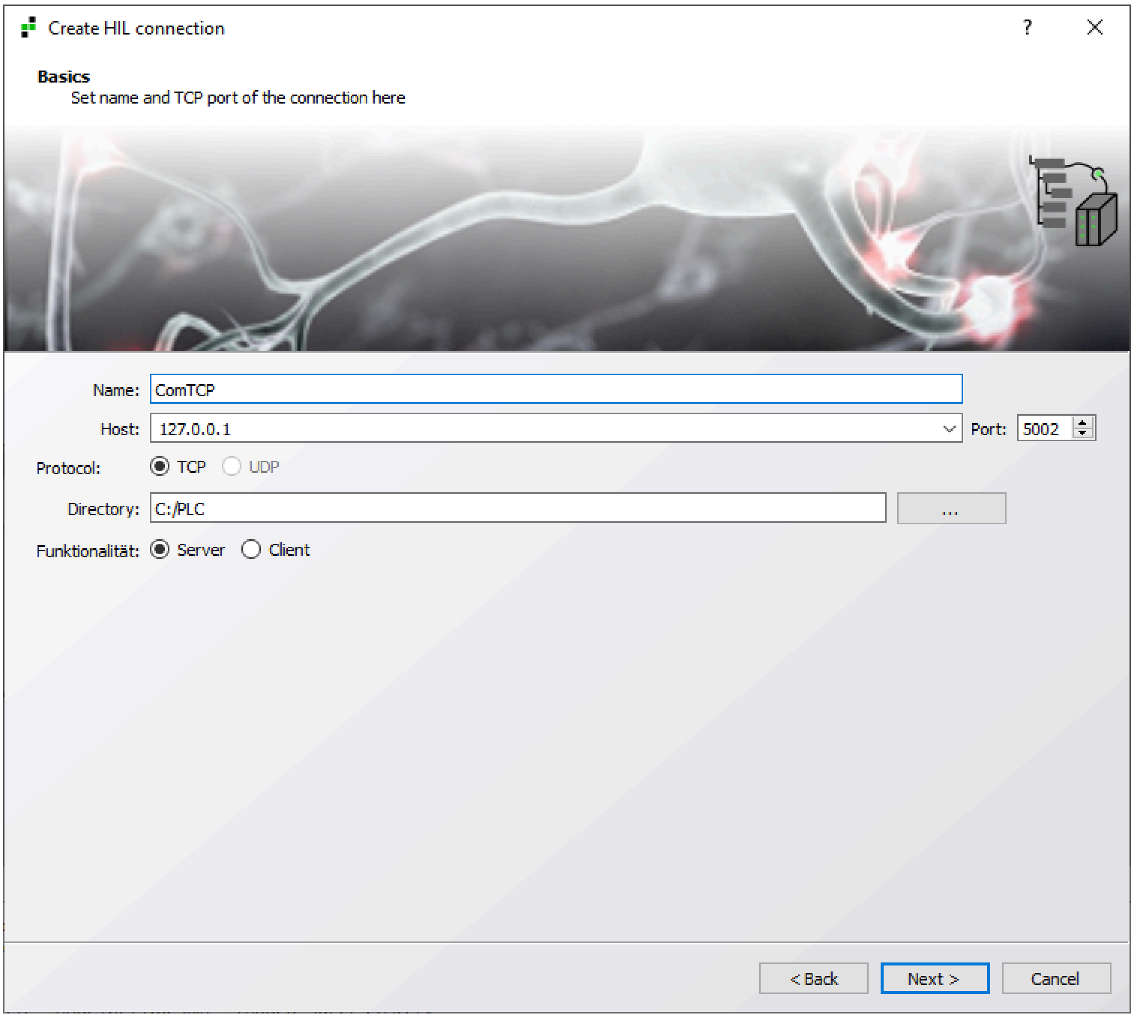


Figure 33: HiL wizard

Labeler

When creating a HiL configuration, only I/Os that are used can be preselected via the labeling editor.

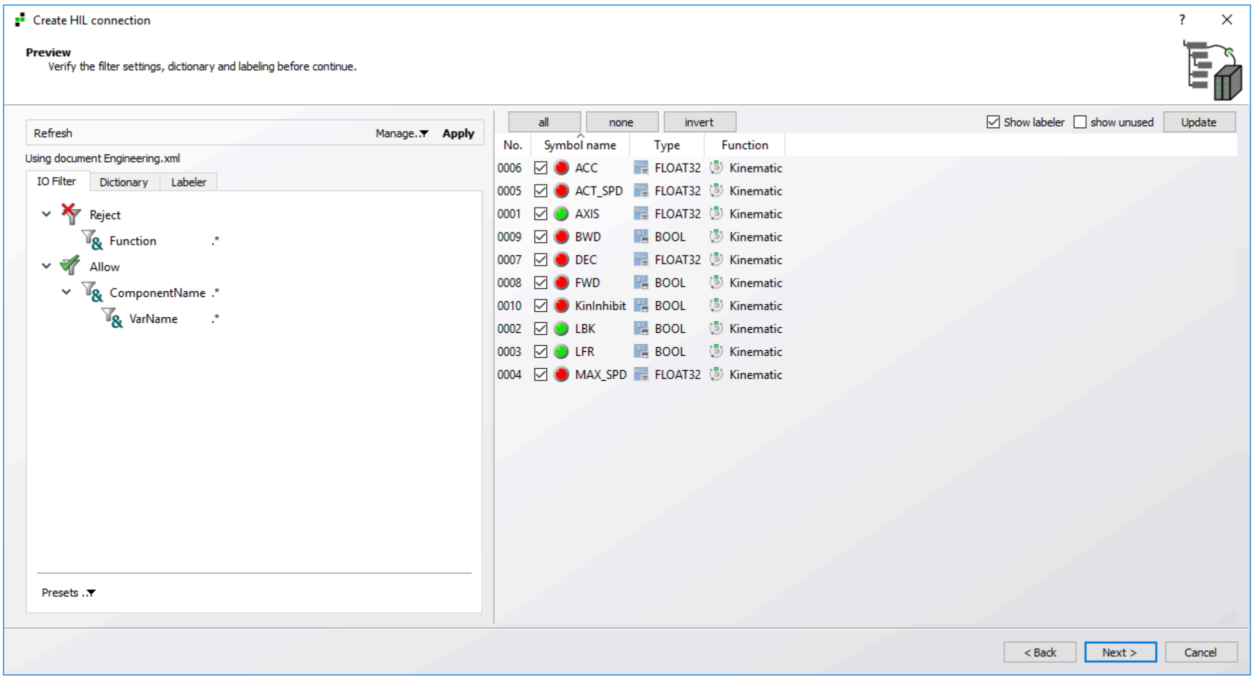


Figure 34: Labeler

It is generally advisable to suppress all variables using a **Reject** filter in order to allow individual variables for individual components.

Communication with Automation Studio

To establish a connection with Automation Studio, the package generated in industrialPhysics must be imported. This means that folder MNG is created in folder "Libraries", which contains a task and a library. Task MNG_Prg should be executed in a 10 ms task class and the library must be added to the project.

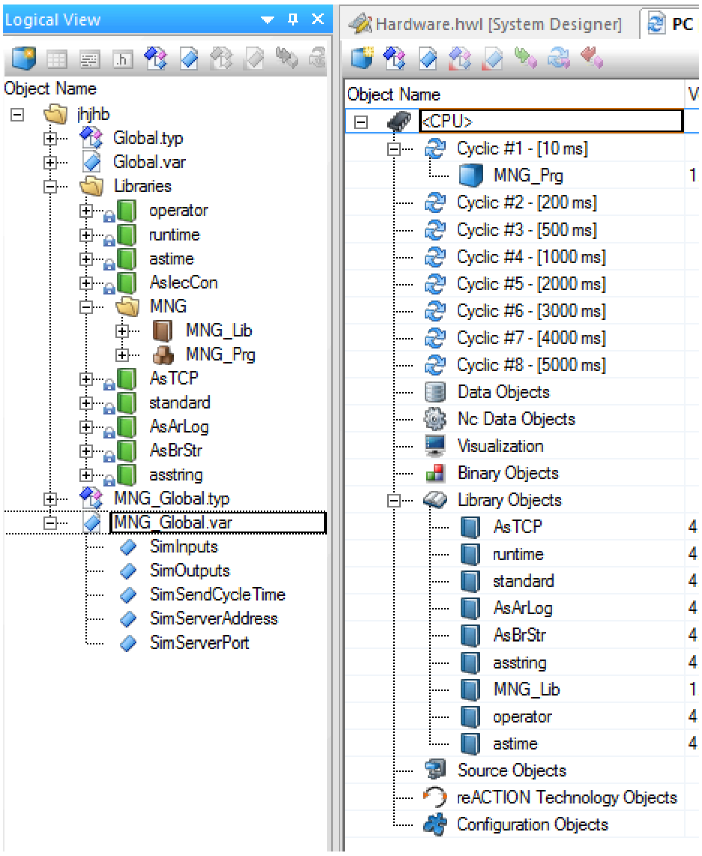


Figure 35: Automation Studio

The communication variables are global structures called "SimInputs" and "SimOutputs".
If the communication package runs on the controller, the node status can be checked in industrialPhysics under **ComTcpView** (Alt + Shift + T or HiL -> ComTcpView).

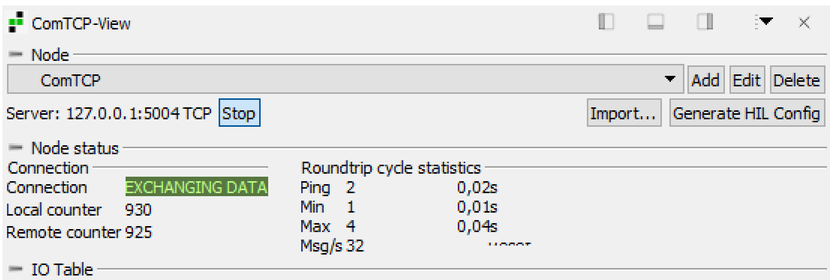


Figure 36: ComTcpView - Node status

?/Knowledge center/doc/40_HIL/BuR/HowTo/Howto_DE_HIL_BuR.pdf

Update HIL connection

If connection settings have been changed or connection variables removed, the **Generate HIL configuration** button can be used to recreate the communication package without having to run the wizard again.
To add new variables, you must run the HIL wizard, however. This is also because the **IO filter** should always be used in the **Label editor**.

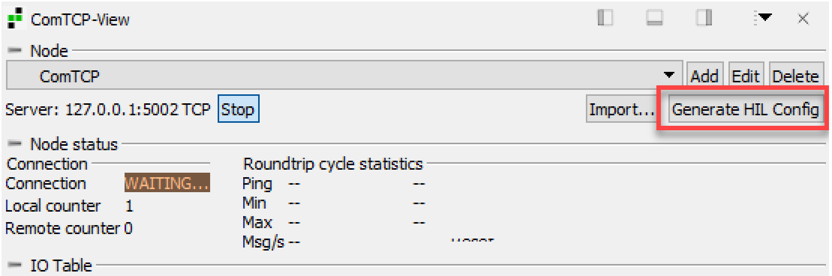


Figure 37: Create HIL configuration

The ZIP file created must then be imported into the Automation Studio project again.

Exercise A: Automation Studio

This section uses the HIL wizard to establish a connection to Automation Studio. In order to control the components with Automation Studio, the TCP connection must be configured via the wizard.

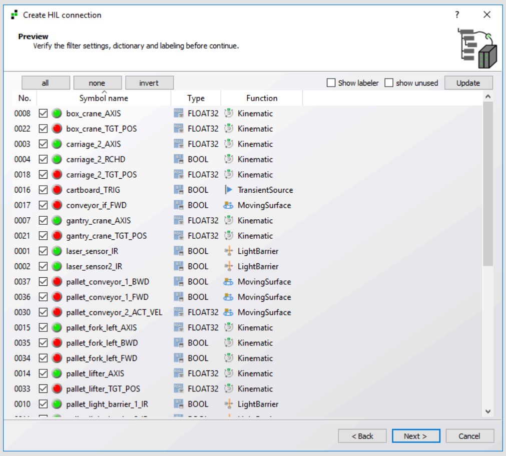


Figure 38: HiL wizard

1) Start the wizard under **HIL** -> **Start wizard**.

2) Select the target platform **BuR Automation Studio 4**.

3) Configure the ComTCP basic settings.

4) Select the desired inputs and outputs.

5) Generate the communication package.

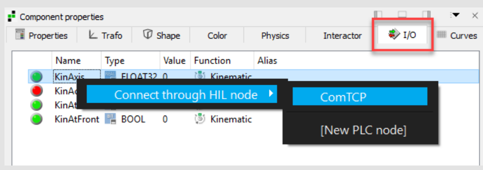
6) Import the ZIP file created into an AS project.

7) Add library **MNG** to your project. (10 ms task)

Exercise B: Automation Studio

In this section, a connection to Automation Studio is established by manually configuring the HIL connection.

In order to control the components with Automation Studio, the TCP connection must be established using the manual configuration possibilities.



1) Create a TCP communication configuration.

2) Configure the basic settings. (Name, Server/Client, IP, Port)

3) Add industrialPhysics variables to this TCP communication configuration.

4) Generate the communication package.

5) Import the ZIP file created into an AS project.

6) Add library **MNG** to your project. (10 ms task)

4.4 Creating new components

Elements

If components are required in addition to those imported from CAD data, they can be added directly in industrial Physics. Apart from the 4 standard shapes cuboid, ball, cylinder and cone, there is a special component called "NoShape". This is an intangible parent node that can be used for grouping real physical components.

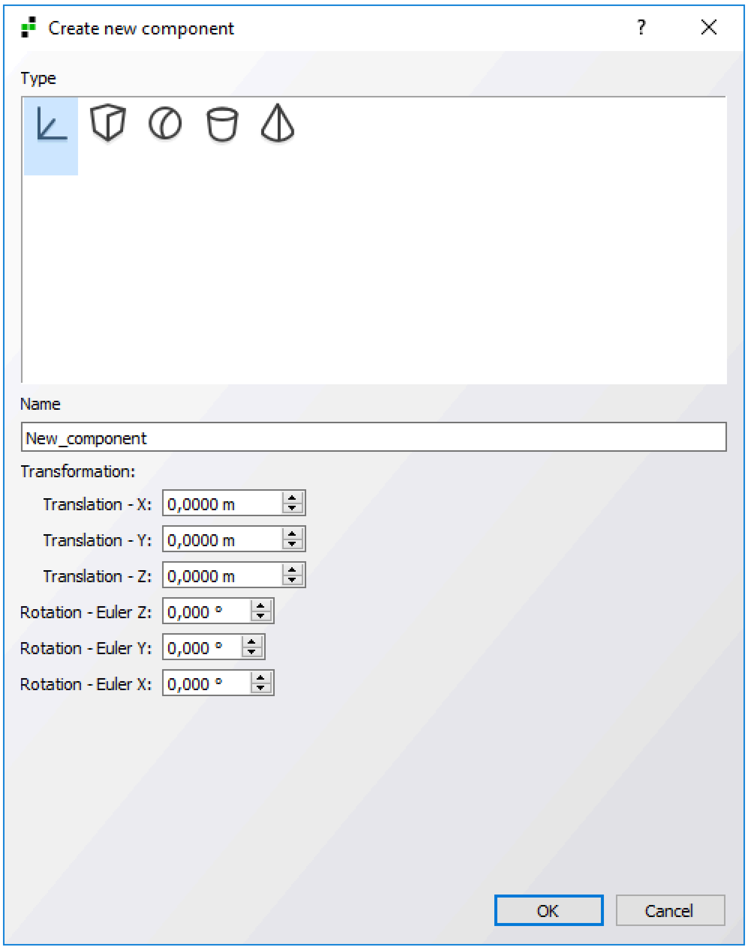


Figure 39: Creating a new component

Position and an alignment can already be specified during creation. If the default settings are kept, however, the object will be created exactly in the center of the coordinate data of the node that is set before the object is added.

Positioning

All components in industrialPhysics can be repositioned at any time. This is done by either holding and dragging the translation arrows or by making changes in window "Component properties".

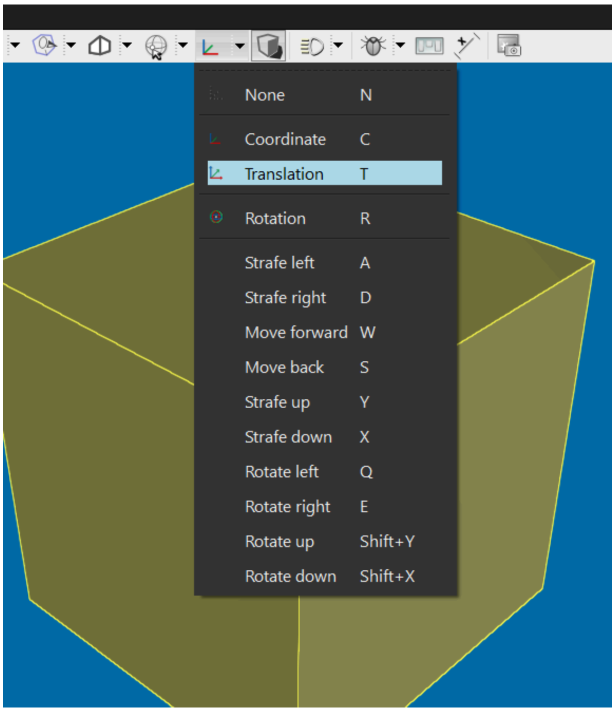


Figure 40: Coordinate

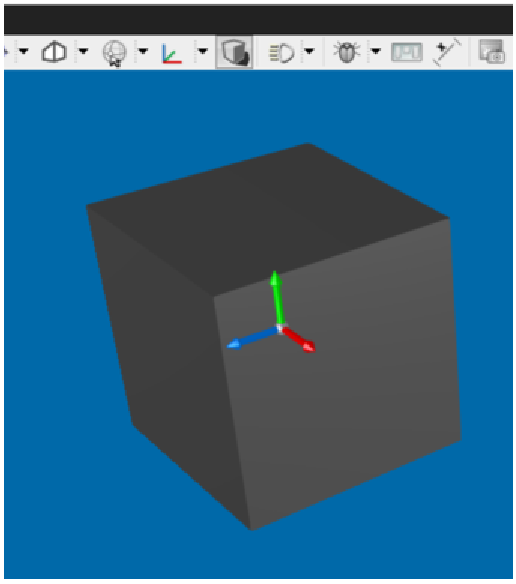


Figure 41: Translation

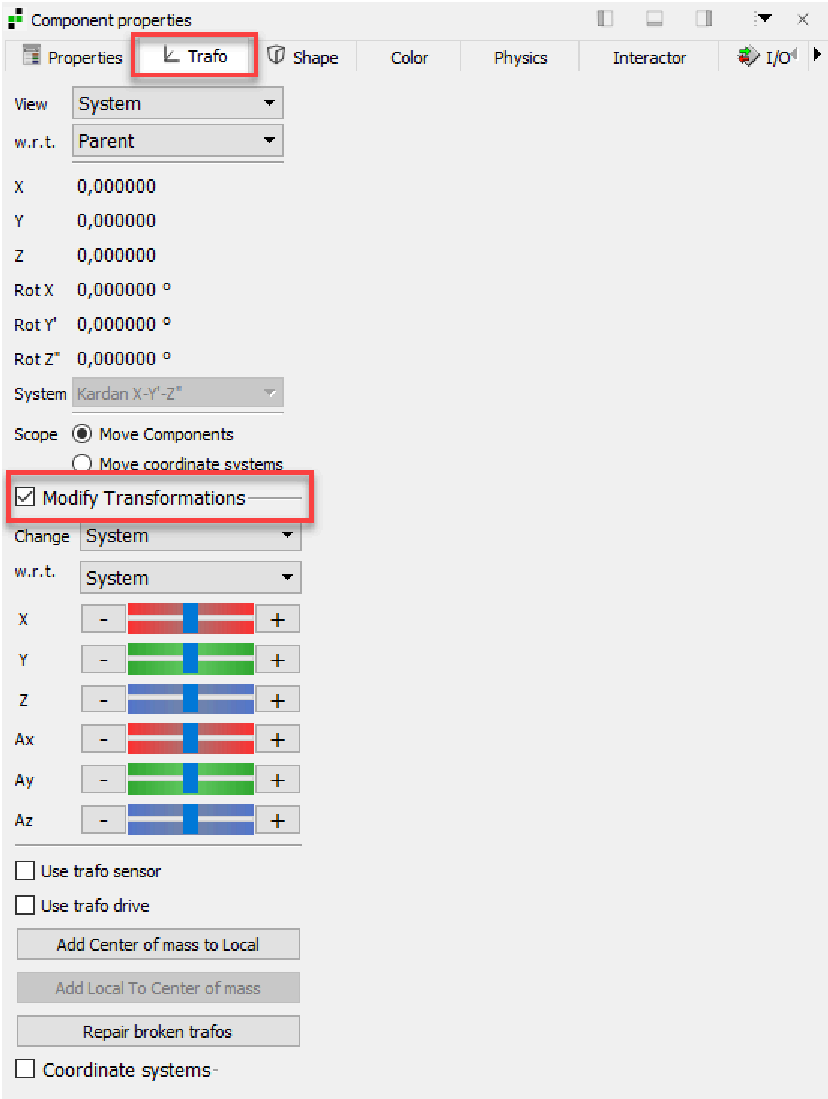


Figure 42: Component properties -> Transformation

Size

The size of elements added to industrialPhysics can also be changed in window "Component properties" under menu option **Shape** if this was not done during creation.

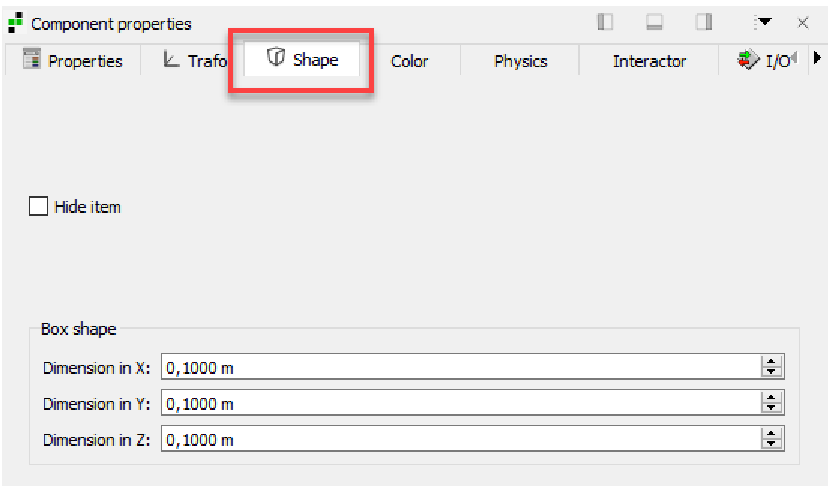


Figure 43: Component properties -> Shape

Task: New components

This section shows how to add new components to a project.
In order to optimize the time sequence for sorting, an additional light barrier is used to detect the correct position of the previous box.

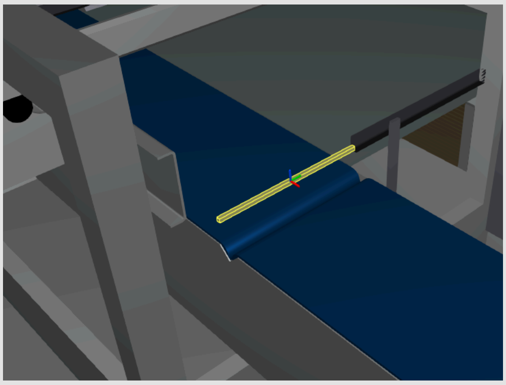


Figure 44: New component: Light barrier

1) Change the property of **Conveyor_Belt_Rotate** and **Conveyor_Belt_End** to **Static** and to **Moving surface**.
2) Adjust the conveyor belt settings. (direction and drive)
3) Change the property of **Belts_Infeed** to **Static**.
4) Add a new box object below node **mb3**.
5) Adjust the new object. (position, size, color)
6) Change the property of the new object to **Static** and **Light barrier**.
7) Test whether the new light barrier detects the correct position of the eighth box.

4.5 Activating the rotary pusher

Friction

industrialPhysics does not differentiate between static and dynamic friction. Each body has a fictitious friction coefficient that can be set under **Physics** in window **Component properties**. In the event of collision between two objects, the resulting friction coefficient is calculated multiplicatively from the two friction coefficients and from the global friction coefficient. The global friction coefficient is defined in the project settings. (Edit -> Properties -> Physics Simulation)

$$F_R \leq \mu_{GLOBAL} * \mu_{Object1} * \mu_{Object2} * F_N = \mu_{\mathrm{Total}} * F_N$$

Recoil

Impact modeling is carried out analogously to the friction by multiplying the impact coefficients of the two components and the global impact coefficient.

$$k_{\mathrm{Total}} = k_{GLOBAL} * k_{Object1} * k_{Object2}$$

$$k = \frac{v'_2 - v'_1}{v_1 - v_2}$$

The number of impacts can also be determined by a drop test.
Based on $v_2=v'_2= 0$, the following applies:

$$k = \sqrt{\frac{h'_1}{h_1}}$$

k = 0: Completely plastic impact
k = 1: Completely elastic impact

?/Knowledge center/doc/10_General/17_Friction/Howto_DE_MOD_FRICTION.pdf

Task: Rotary pusher

This section shows how to make the rotary pusher dynamic in order to fill the pallet truck with the correct number and orientation of boxes.
The pallet should be filled with one row of eight boxes in the vertical direction and two rows of three boxes each in the horizontal direction. In order to rotate the last six packages, the rotary pusher is extended.

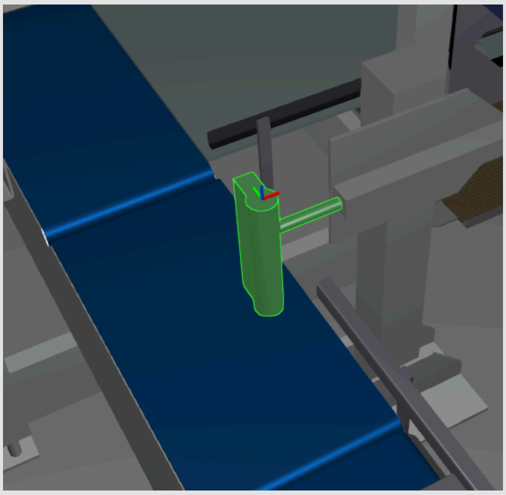


Figure 45: Rotary pusher

1) Change the property of **Rotary pusher<2>** to **Kinematic**.

2) Add **Translation along X** to the **rotary pusher** higher-level element.

3) Change the collision body of **Rotary pusher<2>** to **Cylinder along Z**.

4) Adjust the settings of the **rotary pusher**. **(drive** and **mechanical end stops**)

5) Change the property of the **carriage** to **Kinematic** and **Translation in Y**.

6) Adjust the settings for the **carriage**. **(drive** and **mechanical end stops**)

7) Change the property of **Pallet_Location** to **Static**.

8) Adjust the **friction** and **recoil** of the **rotary pusher**.

9) Create a new ComTCP package and import it into Automation Studio.

10) Adjust your AS project in order to move a complete pallet of boxes onto the crane platform.

4.6 Pallet truck pusher with rails

Curves

If elements should follow more complex paths instead of being moved in simple translational or rotational directions, curves must be defined.
The curve is defined in window "Component properties". To move a component along a curve, the curve must be defined in parent node "NoShape".

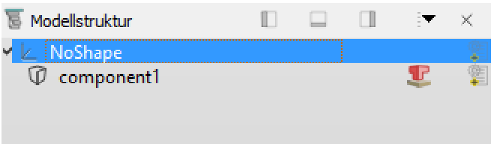


Figure 46: Curves - Model structure

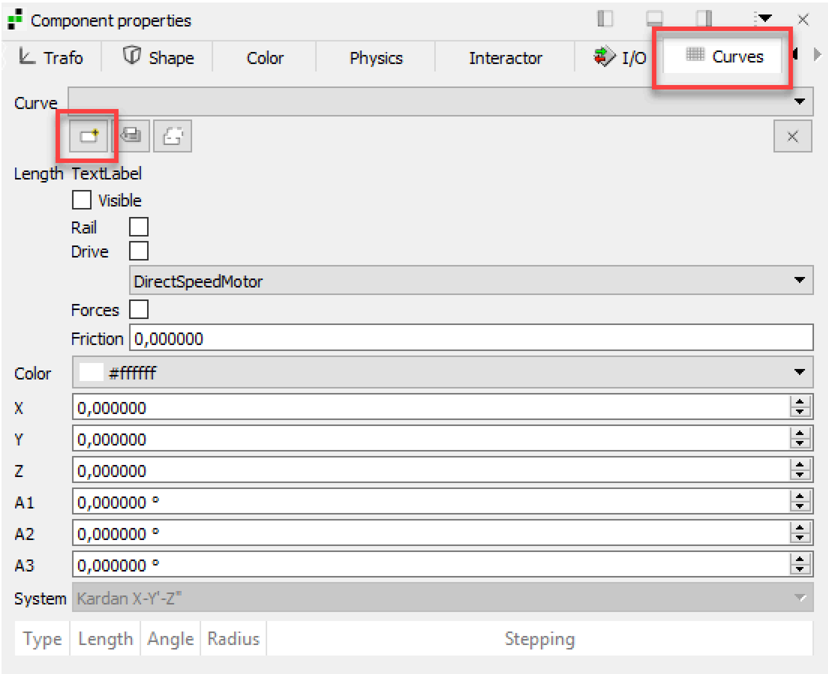


Figure 47: Component properties - Curves

To create a new curve, click on button "Add a new curve". The following settings can be made:

Visibility	Shows or hides the curve.
Rail	Turns the curve into a rail to which dynamic "track cars" adhere automatically.
Drive	A drive can be selected for rails.
Forces	Forces can be activated for rails.
Friction	A friction is defined on the curve.
Color	Defines the color of the curve.
X, Y, Z, Red X, Red Y, Red Z	Transforms the starting point and the alignment of the curve

Table 4: Curve properties

To add individual elements to the curve, right-click in the white field and select the desired object.

When an arc is added, the coordinate system rotates with the arc.

?/Knowledge center/doc/10_General/12_Chain/101_Modeling_chain_finished.iphx

Task: Curve-based pusher

To push the boxes onto the waiting pallet, the pusher must be made dynamic.

The pusher is guided by two belts. In order to perform this movement, a curve must first be defined. The pusher will then follow the movement of the curve.

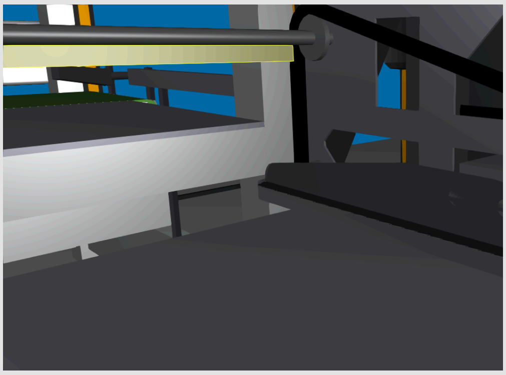


Figure 48: Pallet truck pusher

1) Change the property of the **platform** of the pallet truck to **Kinematic**.

2) Add a new **NoShape object** below node **Portal**.

3) Rearrange elements **Pusher** and **Belt**.

4) Add a new **curve** to node **Belt**.

5) Add individual **curve elements** in order to adjust the path of the belt.

6) Add a **chain** to node **Belt** and configure it.

7) Change the property of the **pusher** to **Kinematic** and test the movement using the chain drive.

8) Change the property of the **portal** to **Translation in X**.

Solution: Rail configuration

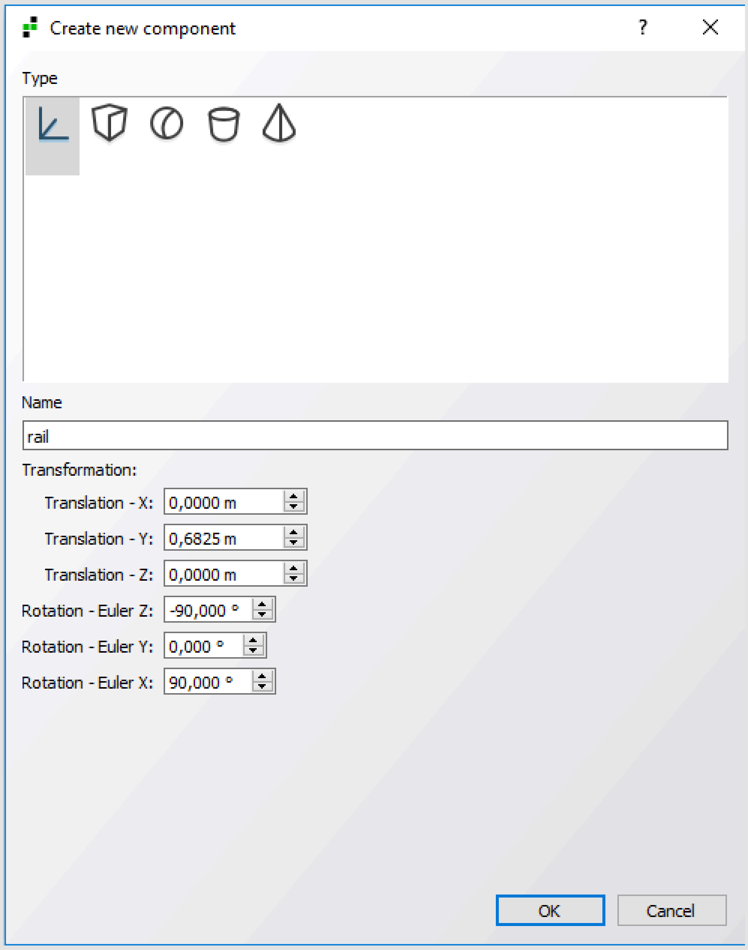


Figure 49: Solution: New NoShape element

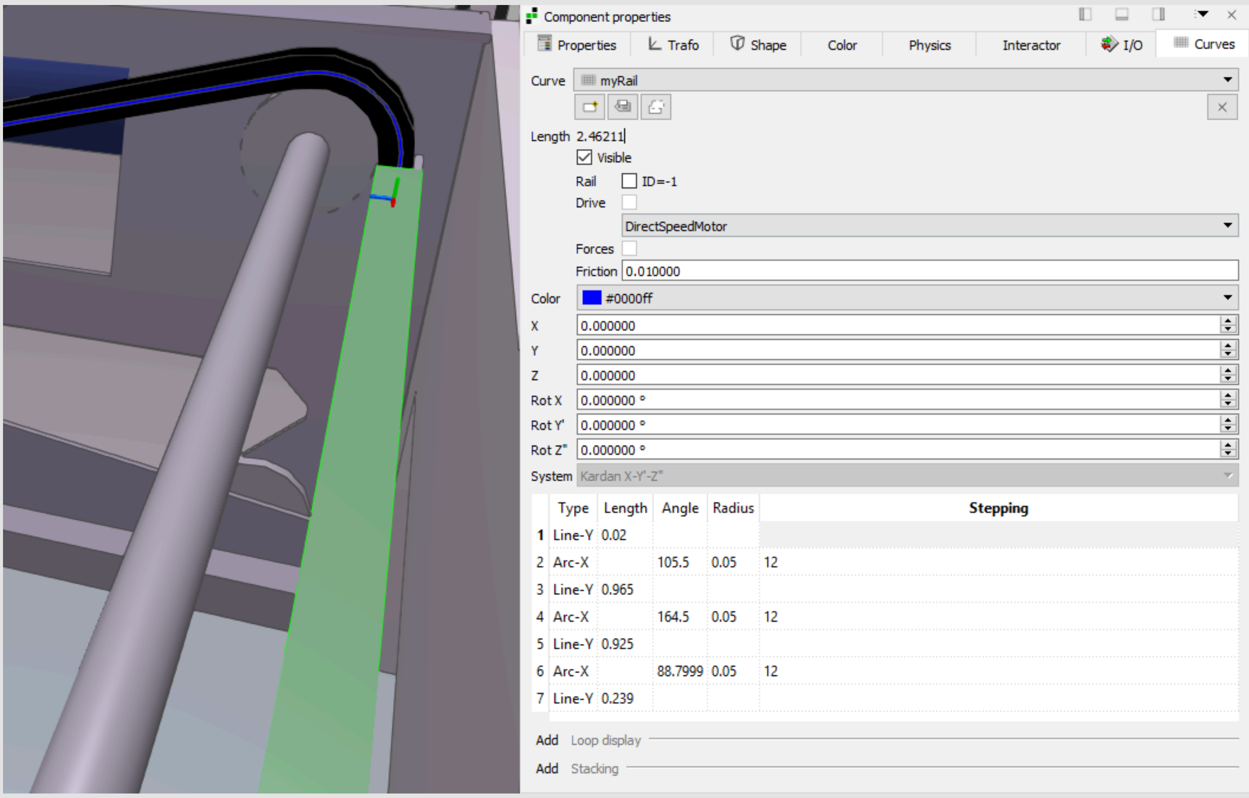


Figure 50: Solution: Rail

4.7 Dynamic gripper with script trigger

Dynamic gripper

Dynamic objects can be detected and moved by **dynamic grippers**. A BOOL input and a BOOL output are used to operate this gripper.
The gripper is switched on via input **Gripper** so that dynamic objects adhere to the gripper object and can thus be moved along or held in place.
Output **Vacuum** provides feedback as to whether the gripper has a dynamic object adhering to it.
There are the following additional setting options:

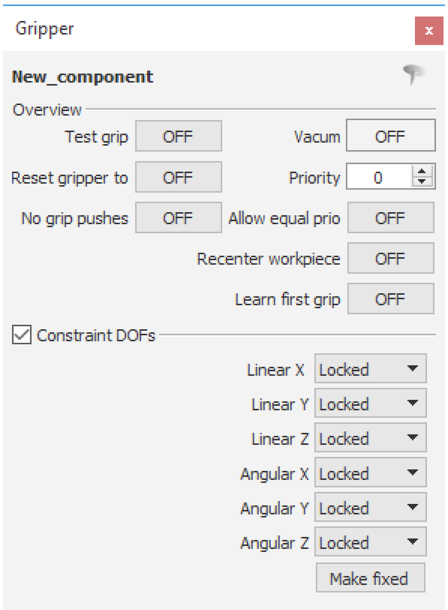


Figure 51: Gripper options

The most important ones are:

- Priority: If several grippers want to grip the same object at the same time, the higher priority determines which gripper the object adheres to.
- Recenter workpiece: Recenter the object on the gripper, regardless of which part of the gripper was used to detect the object.
- Constraint DOFs: The 3 directions and 3 rotations can be released individually so that the object can still move in this direction.

Scripting

It is possible to create scripts to easily automate tasks or to perform processes in industrialPhysics as realistically as possible.
This is done by clicking on button "Insert script" or by double-clicking on the script icon in the model structure.

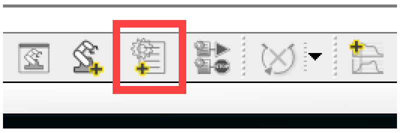


Figure 52: Button "Insert script"

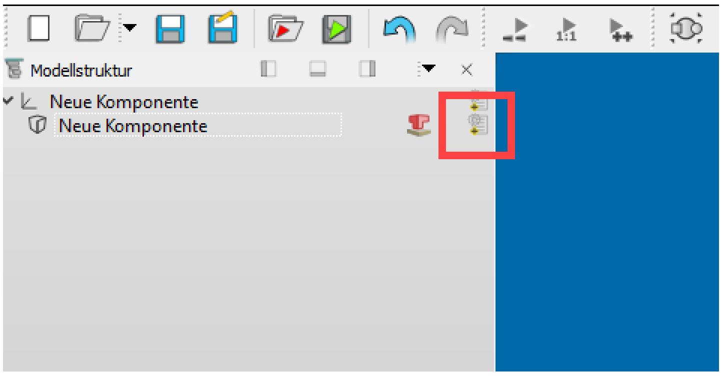


Figure 53: Script icon in model structure

The new script window is empty and thus a script structure must be inserted.

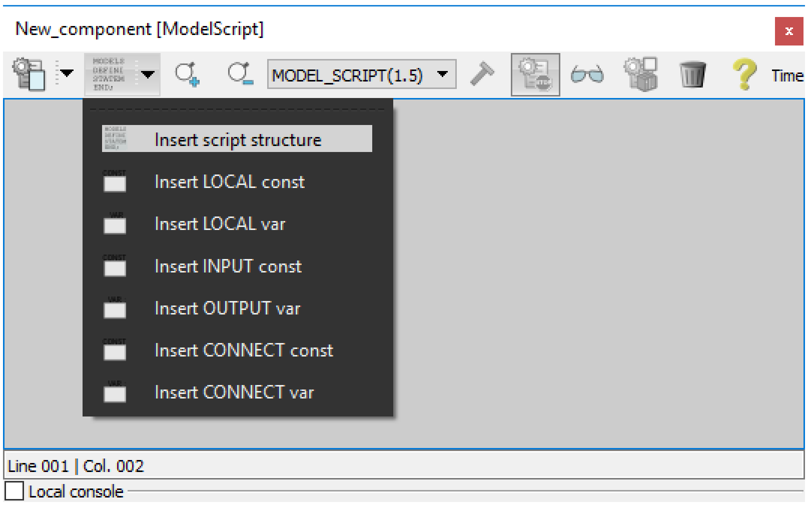


Figure 54: Inserting a strip structure

The same menu option is used to insert templates for the definitions.

Local constants	Constants that can only be used and read locally.
Local variables	Variables that can only be used locally.
Input constants	Constants that can be written to via the TCP connection.
Output variables	Variables that can be transferred to the PLC via the TCP connection.
Connect constants	Constants that can access other model variables.
Connect variables	Variables that can access other model variables.

The connection to other variables can be either a relative or an absolute path specification.
Link a script variable to an existing variable of the same object (with a status variable of a ramped drive in this case):
VAR VarName := CONNECT(".?KinBack");
Link a script variable with an existing variable of an object that is on the same level in the hierarchy but below the same parent node:
VAR VarName := CONNECT("../Sister object?KinBack");
../../ can be linked as desired in order to move on to a higher level in the hierarchy. In some cases, however, it is advisable to use absolute path specifications:
VAR VarName := CONNECT("**Physics://physics/**Node1/Node2/Components?IR");
This path information is also available in window "Component properties" under "Properties":

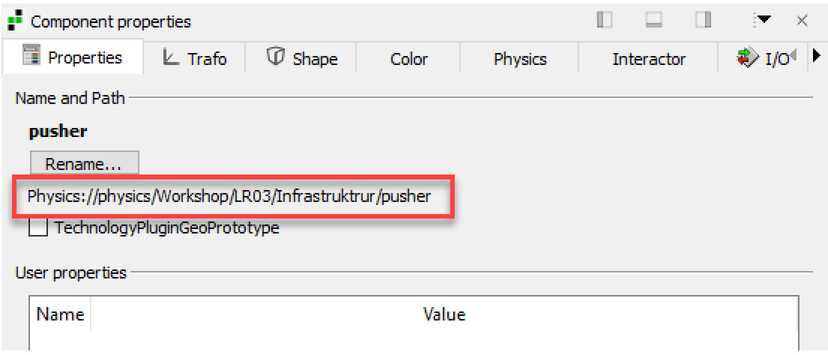


Figure 55: Path information

To save a script, it must be embedded:

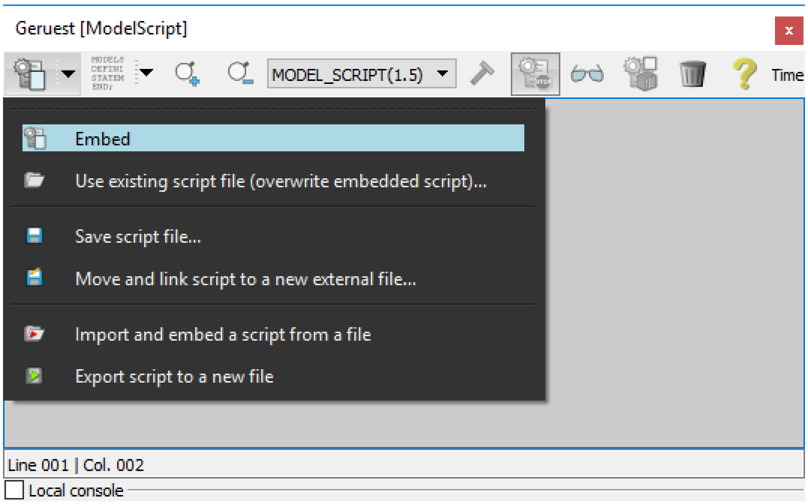


Figure 56: Script - Embed

All keywords and data types can be found in the integrated help documentation:

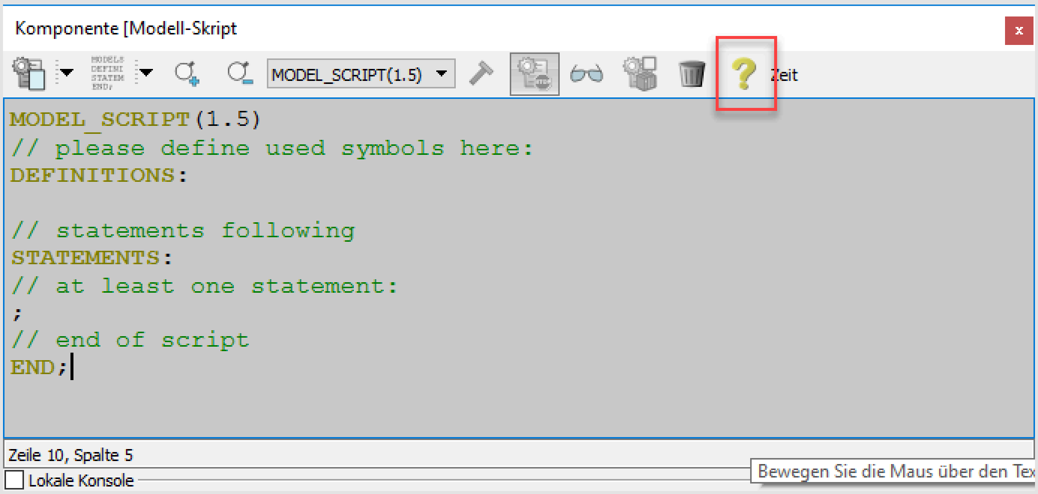


Figure 57: Script - Help documentation

Task: Gripper

An intermediate layer of cardboard is required between two layers of boxes. The intermediate cardboard layer must be collected from a stack by a gripper on the underside of the pallet truck. In order to simulate the automatic feeding of the boxes, a source is created from the top layer and triggered by a script.

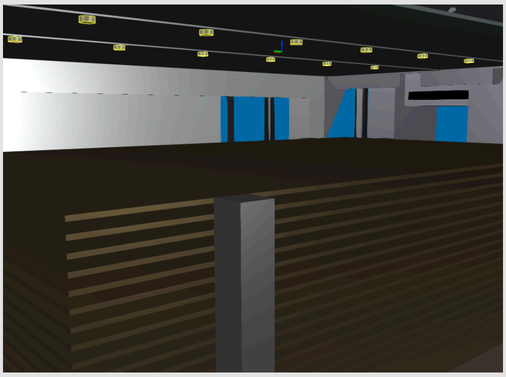


Figure 58: Cardboard boxes

1) Change the property of the **pallet truck** to **Translation in Z**.
2) Adjust the translational properties for the **pallet truck**. (end stops, drive, acceleration and speed)
3) Change the property of **Suction_Pallet truck** to **Kinematic** and to **Gripper (Dynamic)**.
4) Change the property of the top **intermediate layer** to **Static** and create a **source** with **Release by trigger**.
5) Create a script that triggers the creation of an **intermediate layer** by linking the Boolean variable of the trigger with that of the gripper when the **gripper** is **activated**.
6) Extend your AS project by automatically adding an intermediate layer.

4.8 Pallet lift with virtual gripper

Extended use of scripts

In the previous example, two variables were linked to save work steps that only exist in simulation and not in reality. Another example for using a script is when collisions of complex objects would use too much computing power and could therefore not be executed.
It is not possible to let the teeth of the pallet lift reach into the native body of the pallet and thus hold it. This is why a virtual gripper is placed above the pallets. The gripper is switched on and off depending on the horizontal position of the pallet lift. Thus the collision bodies can be modeled as full cuboids.
In addition, only one fork of the pallet lift is moved. Its position is copied by the second fork in order to enable realistic control.

Task: Pallet lift

When a pallet has been filled with 2 layers of boxes, it is transported away and the pallet lift places a new pallet onto the conveyor. The pallet lift required for this is made dynamic in this step.

This example shows how to use a script-based virtual gripper. The physics engine of industrialPhysics reacts unexpectedly at times or otherwise, collisions of native objects would require too much computing power. In this case, several script-controlled workarounds must be implemented in the simulation to avoid any changes to the Automation Studio project. This task aims at adding a virtual gripper for the slightly unstable stack of pallets and avoiding the big CPU load of native object calculations.

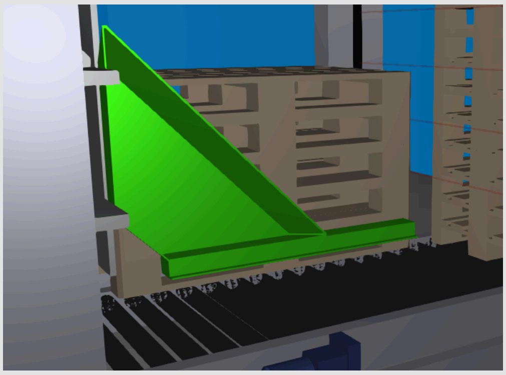


Figure 59: Pallet lift

1) Change the property of **Lifting station PZM2** to **Translation in X**.
2) Adjust the **positioning drive** of the **lifting station**. (end stops, drive, acceleration and speed)
3) Change the property of the **pallet carrier** to **Translation in Z**.
4) Adjust the **drive with ramp** for the **pallet carrier** . (end stops, drive, acceleration and speed)
5) Change the property of **pallet carrier<2>** to **Translatory in Z**.
6) Adjust the **direct drive** for **pallet carrier**<2> . (end stops, drive, acceleration and speed)
7) Link the two pallet carriers with a **script**.
8) Create a new **box element** below node **Lifting station PZM2** and position it.
9) Change the property of the **virtual gripper** to **Kinematic** and **Gripper (Dynamic)**.
10) Create a **script** to control the **virtual gripper**.
11) Change the property of **conveyor rollers 1** to **Static**.
12) Change the property of the **EUR-pallets** to **Dynamic**.
13) Test the functionality.
14) Adjust the Automation Studio project so that a pallet is placed on the conveyor when requested.

4.9 Pallet conveyors

In order to transport the full pallet away and place the empty pallet on the loading position, the conveyor belts must be started.

Exercise: Pallet conveyors

To make the machine complete, the full pallets must be transported away and the new, empty pallets must be moved from the lift to the loading point. For this, the three segments of the conveyor belt must be made dynamic.

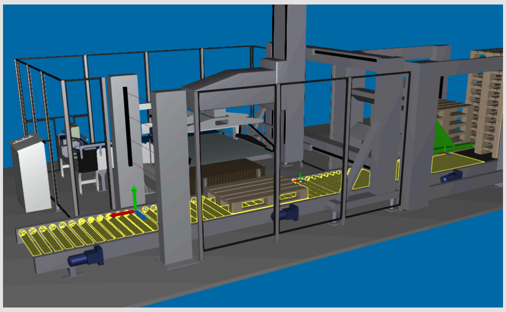


Figure 60: Pallet conveyors

1) Change the collision type of **conveyor rollers 1** to **Moving surface**.
2) Adjust the conveyor. (direction and drive)
3) Change the property of **conveyor rollers 2 and 3** to **Static**.
4) Change the collision type of **conveyor rollers 2 and 3** to **Moving surface**.
5) Adjust the conveyors. (direction and drive)
6) Think about how you can position the empty pallets exactly at the loading position.
7) Adjust your Automation Studio project in order to run the palletizing machine continuously.

5 Summary

Plant simulation and virtual commissioning are the essential aspects of industrialPhysics. Easy to import CAD data and the connection with Automation Studio make it possible to test the application on the digital twin in an efficient way. The connection to industrialPhysics has made it possible to extend simulation in Automation Studio by adding the possibility of simulating plants.

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